

Advanced Machine Learning

DATA 642

Lecture 12 — Introduction to Neural Networks. The Perceptron

*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth, Machine Learning A Bayesian and Optimization Perspective, Sergios Theodoridis, Elsevier, and Hands-On Machine Learning with Scikit-Learn Keras and TensorFlow, Aurelien Geron, O'Reilly

Quick history and facts

- Artificial neural networks (ANNs) were first introduced back in 1943 by the neurophysiologist Warren McCulloch and the mathematician Walter Pitts. In their landmark paper,² "A Logical Calculus of Ideas Immanent in Nervous Activity," McCulloch and Pitts presented a simplified computational model of how biological neurons might work together in animal brains to perform complex computations using propositional logic.
- In the early 1980s there was a revival of interest in connectionism (the study of neural networks), as new architectures were invented and better training techniques were developed. But progress was slow!
- By the 1990s other powerful Machine Learning techniques were invented, such as Support Vector Machines. These techniques seemed to offer better results and stronger theoretical foundations than ANNs, so once again the study of neural networks entered a long winter.
- Geoffrey Hinton + Yoshua Bengio + Yann Lecun received the **2018 Turing Award** for their contributions to the field of neural networks.

- Finally, we are now witnessing yet another wave of interest in ANNs! Why?
 - There is now a huge quantity of data available to train neural networks, and ANNs frequently outperform other ML techniques on very large and complex problems.
 - The tremendous increase in computing power since the 1990s now makes it possible to train large neural networks in a reasonable amount of time. This is in part due to Moore's Law, but also thanks to the gaming industry, which has produced powerful GPU cards by the millions.
 - The training algorithms have been improved.
 - ANNs seem to have entered a virtuous circle of funding and progress. Amazing products based on ANNs regularly make the headline news, which pulls more and more attention and funding toward them, resulting in more and more progress, and even more amazing products.

Some applications of Neural Networks

Image Recognition

Neural networks have achieved remarkable accuracy in tasks like object detection, facial recognition, and image classification.

Natural Language Processing (NLP)

Neural networks power language models capable of tasks like machine translation, sentiment analysis, and text generation.

Medical Diagnosis

Neural networks are used to analyze medical images such as X-rays and MRIs for detecting diseases and abnormalities.

Speech Recognition

Neural networks are behind the accuracy improvements in speech-to-text systems, virtual assistants, and voice-controlled devices.

Drug Discovery

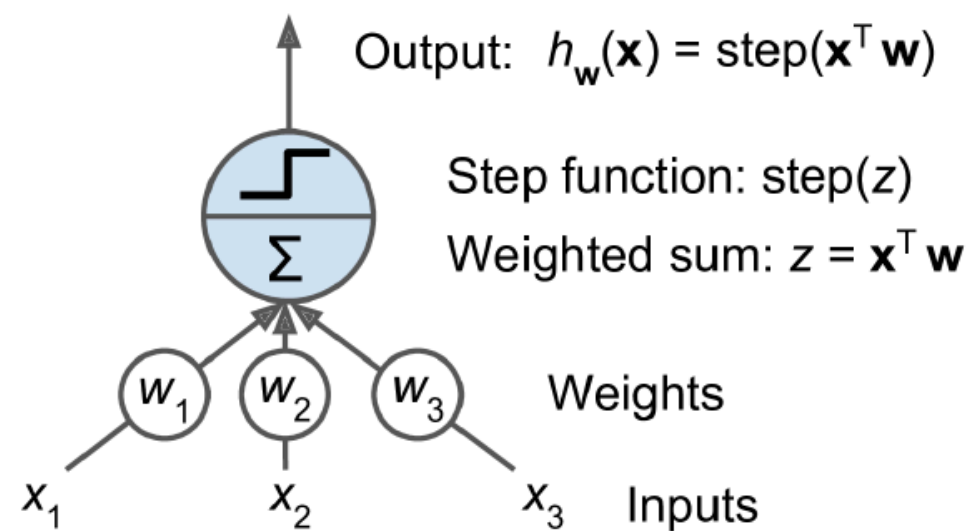
Neural networks aid in identifying potential drug candidates and predicting their properties.

The Perceptron

- The Perceptron is one of the simplest ANN architectures, invented in 1957 by Frank Rosenblatt.
- The Perceptron is based on a threshold logic unit (TLU), or sometimes a linear threshold unit (LTU). The inputs and output are numbers and each input connection is associated with a weight. The TLU computes a weighted sum of its inputs

$$z = w_1x_1 + w_2x_2 + \cdots + w_nx_n = \mathbf{x}^\top \mathbf{w}, \quad (1)$$

then applies a step function to that sum and outputs the result $h_{\mathbf{w}}(\mathbf{x}) = \text{step}(z)$.



The most common step function used in Perceptrons is the Heaviside step function.

- A single TLU can be used for simple linear binary classification. It computes a linear combination of the inputs and if the result exceeds a threshold, it outputs the positive class or else outputs the negative class. Just like a **Logistic Regression** classifier!
- A Perceptron is simply composed of a single layer of TLUs, with each TLU connected to all the inputs.
- When all the neurons in a layer are connected to every neuron in the previous layer (i.e., its input neurons), it is called a fully connected layer or a dense layer.

How is a Perceptron trained? summarized by Siegrid Löwel in this catchy phrase: "Cells that fire together, wire together." The Perceptron is fed one training instance at a time, and for each instance it makes its predictions. For every output neuron that produced a wrong prediction, it reinforces the connection weights from the inputs that would have contributed to the correct prediction.

To mathematically demonstrate this, let $(y_n, \mathbf{x}_n)$ for $n = 1, \dots, N$ be the training set, and let $y_n^{(i)}$ and $\mathbf{x}_n^{(i)}$ be the training pair that is presented in the algorithm at the i th iteration. For a single output layer, the update rule is given by

$$\mathbf{w}^{(i)} = \mathbf{w}^{(i-1)} + \alpha y_n^{(i)} \mathbf{x}_n^{(i)} \quad (2)$$

if $\mathbf{x}_n^{(i)}$ is misclassified by the choice of $\boldsymbol{\theta}^{(i-1)}$, and the update rule is given by

$$\mathbf{w}^{(i)} = \mathbf{w}^{(i-1)} \quad (3)$$

if $\mathbf{x}_n^{(i)}$ is classified correctly. Here, α represents the learning rate.

- The decision boundary of each output neuron is linear, so Perceptrons are incapable of learning complex patterns (similar to Logistic Regression classifiers). However, if the training instances are linearly separable, Rosenblatt demonstrated that this algorithm would converge to a solution. This is called the **Perceptron convergence theorem**.
- Once the Perceptron algorithm has converged, we have the weights, and we can use those to make predictions.

Notes:

- Note that contrary to Logistic Regression classifiers, Perceptrons do not output a class probability; rather, they just make predictions based on a hard threshold. This is one of the good reasons to prefer Logistic Regression over Perceptrons.
- In their 1969 monograph titled Perceptrons, Marvin Minsky and Seymour Papert highlighted a number of serious weaknesses of Perceptrons, in particular the fact that they are incapable of solving some trivial problems like the XOR classification problem.
- However, it turns out that some of the limitations of Perceptrons can be eliminated by stacking multiple Perceptrons. The resulting ANN is called a Multi-Layer Perceptron (MLP).

